Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number:13

Start date:02 Feb 2026

Names: Harsha Vardhan Vinukonda

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Internship Duration: 8 Weeks

1. Project Title

ElectViz Election Data Visualization for Media

2. Project Objective

The primary objective of this project is to analyze the India General Election 2024 dataset and present meaningful insights using data visualization techniques. The project aims to transform large and complex election data into an easy-to-understand format through interactive dashboards developed in Power BI.

Another key objective is to study election results at multiple levels, including **state-level, constituency-level, and party performance analysis**, in order to understand voting patterns and political trends across different regions. The project also focuses on comparing the performance of major alliances such as **NDA, INDIA, and other parties**, providing a clear view of their contribution to the overall election outcome.

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3. Project description

This project focuses on the analysis and visualization of the India General Election 2024 dataset using Power BI. The main aim of the project is to convert raw and large-scale election data into meaningful insights through interactive and visually appealing dashboards. By applying data visualization techniques, the project simplifies complex information and presents it in an easy-to-understand manner.

The project is designed to provide a comprehensive analysis of election results at different levels, including **state analysis, constituency analysis, party performance analysis, and overall election overview**. Each module highlights specific aspects of the election, such as vote share distribution, candidate performance, and party-wise seat allocation.

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Using tools like Power BI, Power Query, and DAX, the dataset is cleaned, transformed, and analyzed to generate accurate and dynamic visualizations. Various charts such as bar charts, donut charts, and gauge charts are used to represent key metrics like total seats, alliance-wise performance, and voting trends.

Overall, this project demonstrates how data analytics and visualization can be effectively used to interpret election data, providing valuable insights for better understanding of electoral outcomes and trends.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Introduction to Data Visualization and Power BI	Successfully set up power BI environment
Week 2	Collection of Dataset	Dataset collected
Week 3	Milestone 1	Milestone 1 Achieved
Week 4	Milestone 2	Milestone 2 Achieved
Week 5	Milestone 3	Milestone 3 Achieved
Week 6	Milestone 4	Milestone 4 Achieved
Week 7	Preparation of PPT and GitHub	Successfully completed GitHub and PPT Creation
Week 8	Mock, senior Mentor connect and final report preparation	Prepared final report and Mock review completed

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Team formation, project topic selection.	02-02-26
Prototype/First Draft	Initial power BI dashboard structure created.	20-02-26
Mid-Term Review	Presented initial dashboard version for feedback.	16-03-26
Final Submission	Completed and submitted fully functional dashboard.	27-03-26
Presentation	Final team presentation showcasing the dashboard.	

5b. Project execution details

The execution of this project was carried out in a structured manner by dividing the work into four major milestones: **State Analysis, Constituency Analysis, Party Performance Analysis, and Overview Analysis**. Each milestone focuses on a specific level of analysis and contributes to the overall understanding of the election data.

Milestone 1: State Analysis

In this phase, the project focuses on analyzing election results at the **state level**. The dataset was filtered and organized to study the performance of political parties across different states. Data cleaning and transformation were performed using Power Query to ensure accuracy and consistency.

Various visualizations were created to represent state-wise insights. A **donut chart** was used to display vote share distribution among parties within a selected state. Bar charts were implemented to highlight top-performing constituencies based on total votes. Additionally, slicers were introduced to allow users to dynamically select and analyze specific states.

This milestone helps in identifying **dominant parties in each state**, understanding regional voting patterns, and comparing party performance within states. It provides a clear picture of how election results vary across different regions.

Milestone 2: Constituency Analysis

This milestone provides a detailed analysis at the **constituency level**. The focus was on understanding candidate performance and competition within each constituency. The dataset was structured to include key attributes such as candidate names, party affiliation, total votes, and winning status.

Bar charts were used to display the **top candidates based on votes**, while donut charts represented party-wise vote share within a constituency. Measures were created using DAX to calculate metrics such as total candidates and vote distribution.

A comparison between the **winner and runner-up** was also included to analyze winning margins and competitiveness. Interactive filters were added to allow users to select a specific constituency and explore detailed insights.

This milestone helps in understanding **local-level competition, voter preferences, and candidate performance** in each constituency.

Milestone 3: Party Performance Analysis

In this phase, the project focuses on evaluating the performance of different political parties. The dataset was grouped based on party names to calculate the number of seats won and overall contribution to alliances.

Bar charts were used to represent **seats won by each party**, enabling easy comparison among major political parties. Additional analysis was done to understand the role of parties within alliances such as **NDA, INDIA, and Other parties**.

DAX measures were created to calculate party-wise seat counts and rankings. This allowed identification of **top-performing parties** and their impact on the overall election results.

This milestone provides insights into **which parties performed best, their strengths, and their contribution to forming the government**.

Milestone 4: Overview Analysis

The final milestone presents a **comprehensive summary of the entire election dataset**. This dashboard combines insights from all previous milestones to provide a high-level view of election results.

Key Performance Indicators (KPIs) were created using cards to display **NDA seats, INDIA seats, and Other seats**. A **donut chart** was used to visualize alliance-wise seat distribution, while a **bar chart** showed seats won by each party.

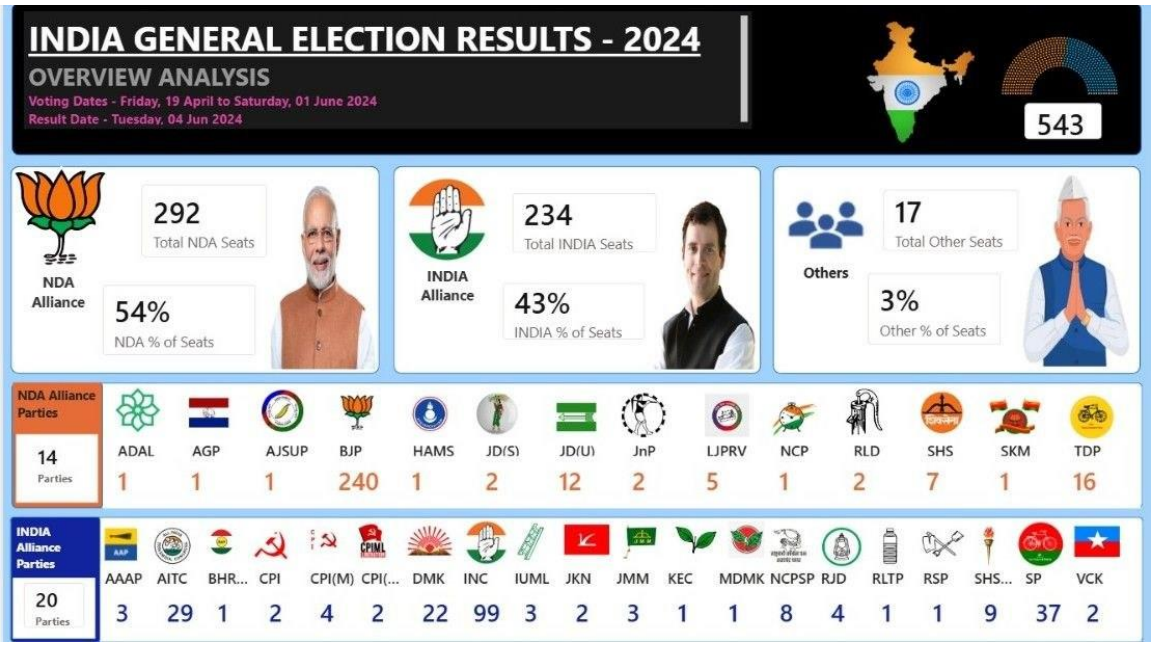
A **gauge chart** was implemented to represent the total number of seats (543), indicating the overall scale of the election. Interactive filters were included to allow users to explore the data based on different parameters such as state and party.

This milestone provides a **complete overview of election outcomes**, helping users quickly understand the overall performance of alliances and parties.

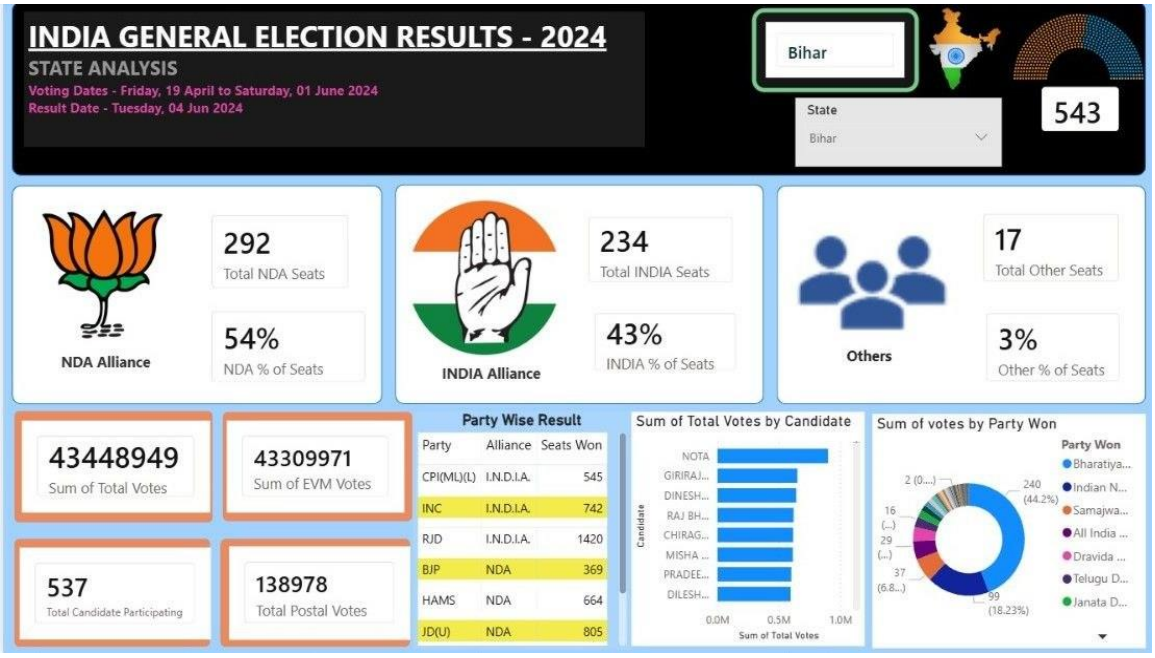
The project execution followed a step-by-step approach, starting from data preparation to advanced visualization. Each milestone builds upon the previous one, ensuring a **comprehensive and multi-level analysis of election data**. The use of Power BI tools and techniques enabled the creation of **interactive, dynamic, and insightful dashboards**.

6. Snapshots / Screenshots

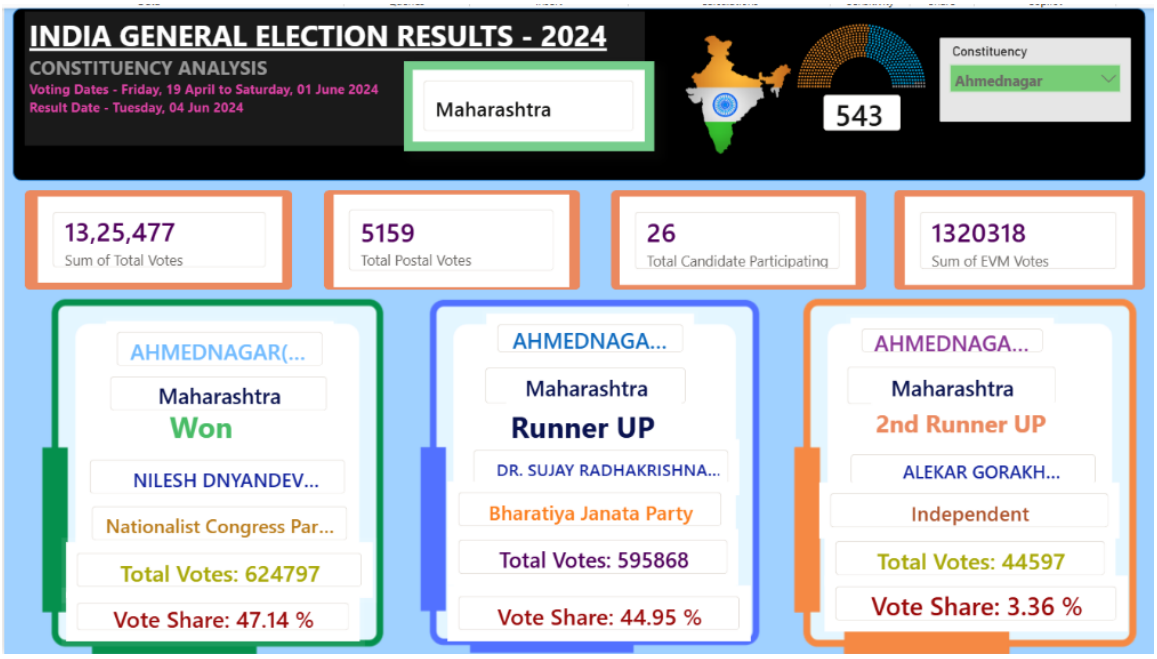
1.Overview Analysis



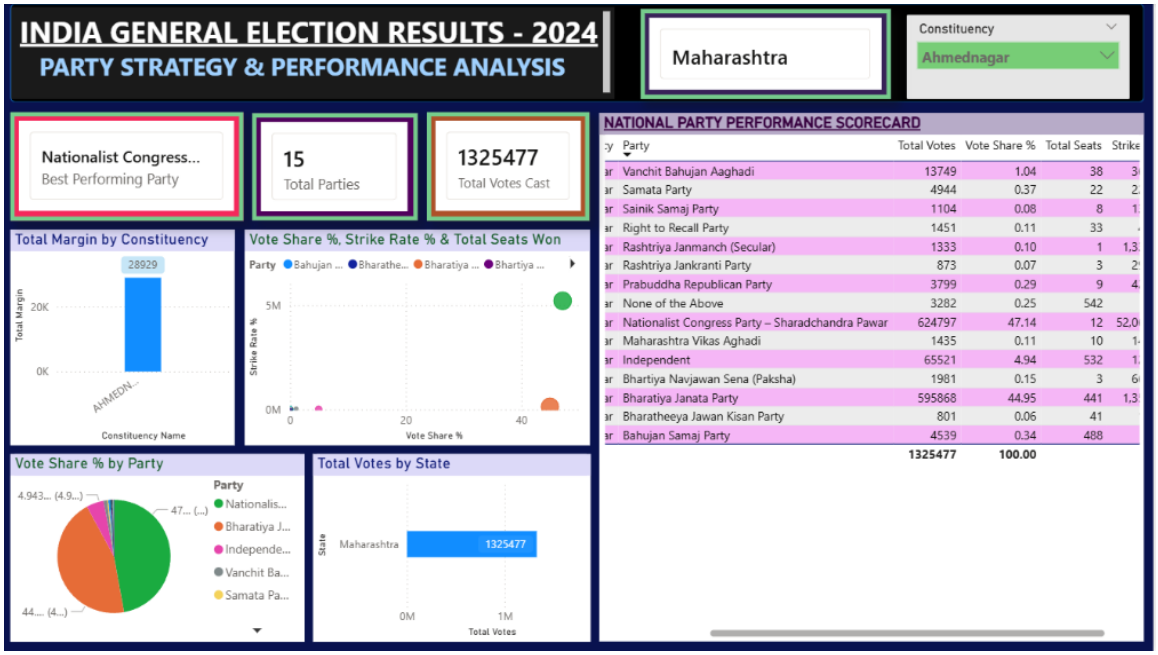
2.State Analysis



3.Constituency Analysis



4.Party Strategy & Performance Analysis



7. Challenges Faced

During the development of this project, several challenges were encountered at different stages. These challenges were addressed through appropriate solutions to ensure successful completion of the project.

Challenge 1: Data Cleaning and Inconsistency

Challenge:
The dataset contained inconsistencies such as missing values, incorrect party names (e.g., NOTA shown as "None of the Above"), and unstructured data, which affected the accuracy of analysis.

Resolution:
Data cleaning was performed using **Power Query Editor**, where missing values were handled, unnecessary columns were removed, and party names were standardized. This ensured consistency and improved data quality for further analysis.

Challenge 2: Creating Accurate DAX Measures

Challenge:
While creating measures like total seats, alliance-wise seats (NDA, INDIA, Others), and vote calculations, incorrect formulas initially resulted in blank or incorrect outputs.

Resolution:
DAX formulas were carefully reviewed and corrected using appropriate functions such as **CALCULATE, COUNT, and SUM**. Filters and conditions were properly applied to ensure accurate calculations.

Challenge 3: Visualization Errors in Power BI

Challenge:
Issues were faced while creating charts, such as:

- “Non-measure field required” errors
- Automatic count aggregation when dragging fields
- Incorrect axis assignments

Resolution:
Fields were properly categorized as measures or columns, and aggregation settings were adjusted. Correct fields were assigned to X-axis and Y-axis, ensuring proper visualization of data.

Challenge 4: Alliance Classification

Challenge:
Categorizing parties into alliances (NDA, INDIA, Others) was difficult because the dataset did not contain a predefined alliance column.

Resolution:
A new **calculated column** was created to manually map parties into their respective alliances using conditional logic. This enabled accurate alliance-wise analysis.

Challenge 5: GitHub Repository Management

Challenge:
Understanding how to structure the repository, add collaborators, and manage contributors was initially confusing.

Resolution:
The repository was organized into folders such as dataset, dashboard, and screenshots. Team members were added as collaborators, and each member made commits to appear in the contributors section.

Challenge 6: Power BI Access Issues

Challenge:
Frequent sign-in requests and restrictions with personal accounts caused interruptions while working in Power BI.

Resolution:
The project was continued using available access, and work was saved locally. Screenshots and exported files were used to ensure smooth project submission.

Despite facing multiple challenges during data preparation, visualization, and repository management, appropriate solutions were implemented at each stage. These challenges helped improve problem-solving skills and provided practical experience in handling real-world data analysis tasks.

8. Learnings & Skills Acquired

During the development of this project, we gained valuable knowledge and practical experience in both **technical and soft skills**. The project provided an opportunity to apply theoretical concepts to real-world data and enhanced our overall understanding of data analytics and visualization.

Technical Skills

Throughout the project, we developed strong technical skills related to data analysis and visualization. We learned how to use **Power BI** effectively to create interactive dashboards and meaningful visualizations.

We gained hands-on experience in **data cleaning and transformation** using Power Query Editor, which helped us handle missing values, remove inconsistencies, and prepare structured data for analysis.

We also improved our knowledge of **DAX (Data Analysis Expressions)** by creating custom measures for calculating total votes, seats, and alliance-wise performance. This helped us understand how to perform advanced calculations and derive insights from data.

Additionally, we learned how to design different types of visualizations such as **bar charts, donut charts, and gauge charts**, and how to use filters and slicers to make dashboards interactive and user-friendly.

Analytical Skills

This project enhanced our ability to analyze large datasets and extract meaningful insights. We learned how to interpret election data at different levels, including **state-level, constituency-level, and party-level analysis**.

We also developed skills in **data interpretation and trend analysis**, enabling us to identify patterns such as party dominance, voting behavior, and performance comparison between alliances.

Soft Skills

Working on this project helped us improve important soft skills such as **team collaboration, communication, and time management**. We worked together by dividing tasks and coordinating effectively to complete the project within the given timeline.

We also improved our **problem-solving skills** by overcoming challenges related to data inconsistencies, visualization errors, and tool limitations.

Furthermore, we developed **presentation and data storytelling skills**, learning how to explain complex data in a simple and understandable manner.

9. Testimonials from team

Working on this project was a valuable learning experience for our entire team. It provided us with practical exposure to **data analysis and visualization using Power BI**, helping us understand how to transform raw election data into meaningful insights through interactive dashboards.

Throughout the project, we gained hands-on experience in **data cleaning, transformation, and creating DAX measures**, which enhanced our technical knowledge. We also developed strong **analytical skills** by interpreting election data at state, constituency, and party levels to identify trends and patterns.

In addition to technical skills, this project helped us improve our **teamwork, communication, and problem-solving abilities**. By dividing tasks and collaborating effectively, we were able to complete the project within the given timeline.

10. Conclusion

In conclusion, this project successfully demonstrates the use of **data analytics and visualization techniques** to analyze the **Indian General Election 2024 dataset**. By using Power BI, we were able to transform complex and large-scale data into **interactive and meaningful dashboards** that provide clear insights into election results.

The project covered multiple levels of analysis, including **state analysis, constituency analysis, party performance, and overall overview**, which helped in understanding voting patterns, party dominance, and election outcomes. The use of various visualizations such as bar charts, donut charts, and gauge charts made the data easy to interpret and compare.

Additionally, this project enhanced our ability to work with real-world data, apply analytical thinking, and present insights effectively. It also helped us develop both technical and teamwork skills.

Overall, the project highlights the importance of **data-driven decision making** and shows how visualization tools can be used to simplify complex information and support better understanding of election data.

11. Acknowledgements

We would like to express our sincere gratitude to **Infosys Springboard Virtual Internship 6.0** for providing us with the opportunity to work on this project and gain practical knowledge in the field of data analytics and visualization.

We are especially thankful to our mentor **Mr. Ajay V**, for his continuous support, valuable suggestions, and guidance throughout the project. His encouragement helped us successfully complete this work.

We would also like to acknowledge cooperation and teamwork of all our team members whose efforts made successful completion of project. Each members's contribution were instrumental in turning ideas into impactful visual insights.

Finally, we are grateful for this enriching experience, which has strengthened our understanding of data analytics and inspired us to pursue further learning and innovation in this field.